

IFN647 – Assignment 2

Specification and Requirements

The methodology you will use for Assignment 2 includes the following tasks:

- **Task 1 – Baseline Models:** Extend the BM25-based IR model and language models to rank documents in each dataset according to the corresponding topic.
- **Task 2 – Custom Model:** Design your own ranking model leveraging knowledge from this unit, including learning-to-rank and pseudo relevance techniques, to rank documents in each dataset.
- **Task 3 – Implementation:** Implement the three models in Python and test them on the provided datasets for their respective topics.
- **Task 4 – Evaluation and Validation:** Evaluate all three models using three standard effectiveness measures to provide a comprehensive comparison of ranking performance. Clearly report and analyse the results for each model across all datasets. In addition, describe the validation strategy used to tune the parameters of your custom model.
- **Task 5 – Significance Testing:** Conduct statistical tests to determine whether your custom model outperforms the two baseline models.
- **Task 6 – Application Scenario and future work:** Describe a potential application scenario. Discuss how your model could be extended to improve performance in the future and consider any relevant ethical issues.

Assignment 2 Data Collection

This data collection is a subset of the RCV1 data collection. It is provided exclusively for IFN647 students supervised by the unit coordinator. Due to copyright and privacy constraints, this data collection must not be shared or distributed to external parties.

- **“Topics.txt”** – This file contains the definitions of all topics. Each `<Topic>...</Topic>` element specifies a topic, including the topic number (`<num>`, e.g., R101), title (`<title>`), description (`<desc>`), and narrative (`<narr>`).
- **“Doc_Collection.zip”** – This archive contains multiple document datasets (folders “Dataset101” to “Dataset150”), where each dataset corresponds to a specific topic. For example, *Dataset101* is associated with topic R101.
- **“Relevant_Judgements.zip”** – This archive contains relevance judgements for all topics. For instance, the file “Dataset135.txt” provides the ground-truth relevance labels for *Dataset135*. In this file, a value of “1” in the third column indicates that the document (second column) is relevant to topic R135 (first column), while “0” indicates that the document is non-relevant.

Assignment 2 Task Specification

Task 1: Design two baseline models, **Baseline1** and **Baseline2**, by extending the BM25-based information retrieval model and the Jelinek–Mercer (JM) smoothing language model, respectively. Both models must operate using the same inputs and produce outputs in a consistent format.

Inputs:

- All topics provided in *Topics.txt*
- All datasets contained in *Doc_Collection.zip*

Outputs:

- Ranked document lists from the databases are saved as text files for all topics.
 - For example, for topic *R135*:
 - Baseline 1 → *Baseline1_R135_Ranking.dat*
 - Baseline 2 → *Baseline2_R135_Ranking.dat*
- All output files must be saved in the folder *ModelOutputs*

You are required to apply:

- Equation 1 for Baseline 1 (BM25-based model)
- Equation 2 for Baseline 2 (JM smoothing language model)

For each topic i , use its title as a query Q_i to compute the **BM25 score** (Baseline 1) and the **JM-smoothed likelihood ranking score** (Baseline 2) for every document D in the corresponding dataset $Dataset_i$, using the specified equations.

$$BM25(D, Q_i) = \sum_{t \in Q_i} \log_{10} \left(1 + \frac{N - n_t + 0.5}{n_t + 0.5} \right) \cdot \frac{(k_1 + 1)f_t}{K + f_t} \cdot \frac{(k_2 + 1)qf_t}{k_2 + qf_t} \quad (1)$$

where $k_1 = 1.2$, $k_2 = 500$, $b = 0.75$, $K = k_1 * ((1 - b) + b * dl / avdl)$, dl is document D 's length and $avdl$ is the average length of a document in the dataset $Dataset_i$.

$$JM(D, Q_i) = \sum_{j=1}^n \log_{10} \left((1 - \lambda) \frac{f_{q_j, D}}{|D|} + \lambda \frac{c_{q_j}}{|Dataset_i|} \right) \quad (2)$$

where $f_{q_j, D}$ is the number of times query word q_j occurs in document D , $|D|$ is the number of word occurrences in D , $|Dataset_i|$ is the total number of word occurrences in dataset $Dataset_i$, and parameter $\lambda = 0.3$.

- (a) Formally describe the design of **Baseline 1** as an algorithm that ranks documents in each dataset using the corresponding topic, applied across all datasets. Your description must clearly define the meaning of the parameters N , n_t , f_t and qf_t within the algorithm.
- (b) Formally describe the design of **Baseline 2** as an algorithm that ranks documents in each dataset using the corresponding topic, applied across all datasets. Your

description must clearly define the meaning of the parameters n and c_{qj} within the algorithm.

Hint: You may begin by designing an outer loop that iterates over all datasets. In each iteration, specify the steps required to process a single dataset and produce the corresponding ranked output.

Task 2. Design your own ranking model, **Model_C**, leveraging knowledge from this unit, such as learning to rank, and pseudo relevance techniques, to rank documents in each dataset.

Model_C must operate using the following inputs and produce outputs in a consistent format as you did for task 1.

Inputs:

- All topics provided in *Topics.txt*
- All datasets contained in *Doc_Collection.zip*

Outputs:

- Ranked document lists from the databases are saved as text files for all topics.
 - For example, for topic *R135*:
 - $\text{Model_C} \rightarrow \text{ModelC_R135_Ranking.dat}$
- All output files must be saved in the folder *ModelOutputs*

Formally describe your design for **Model_C** in an algorithm to rank documents in each dataset using the corresponding topic for all datasets. Your approach should be generic that means it is feasible to be used for other topics. You also need to discuss the differences between **Model_C** and the other two baseline models (**Baseline1** and **Baseline2**).

Task 3. Use Python to implement three models: **Model_C**, **Baseline1** and **Baseline2**, and test them on the given all datasets for the corresponding topics.

Design Python programs to implement these three models. You can use a .py file (or an .ipynb file) for each model. Describe the Python package or module (or any open-source software) you used; and the data structures used to represent a single document and a set of documents for each model (you can use different data structures for different models).

For each topic, your python programs will produce ranked results and save them into .dat files. For example, for topic *R135*, you save the ranked results of three models (**Baseline1**, **Baseline2** and **Model_C**) into “Baseline1_R135_Ranking.dat”, “Baseline2_R135_Ranking.dat”, and “ModelC_R135_Ranking.dat”, respectively by using the following format, where the first column is the document *ID* (the *itemid* in the corresponding XML document in Dataset135) and the second column is the document score (or probability).

Example ranking in Baseline2_R135_Ranking.dat:
Query is the topic title = “WTO trade debates”

Doc_ID	JM_Score
65465	-6.789653953902333
66118	-6.789653953902333
78305	-6.92865409501318
68756	-7.020521602539402
80891	-7.053671763723248
31784	-7.105151033064315
65311	-7.10816230395036
65501	-7.111453815463792
52721	-7.134382342489755
79698	-7.17891940255064
78998	-7.17891940255064
86691	-7.23708047455695
67433	-7.247359939349815
67275	-7.247359939349815
34733	-7.266560364392295
49229	-7.315529960963838
46122	-7.315529960963838
7741	-7.316933938396067
82689	-7.422644373876908
59518	-7.42804492493483
66181	-7.449924695338548
44221	-7.501096634912675
41556	-7.722121800099984
40131	-7.722121800099984
57876	-8.044909540958411
71894	-8.179434479837782
...	

You also need to test the three models on all datasets for the corresponding topics by printing out the top 10 documents for each dataset (in descending order). The output should also be put in an appendix of your final report.

The following are examples of some outputs of Baseline1.

```
R101 (Doc_ID BM25_Score) :
46974 1.8878533347166448
46547 1.8878533347166448
62325 1.4153502848719022
6146 0.8552738202464256
61329 0.8018094461466142
22170 0.7463997606217948
61780 0.6405403133414161
22513 0.5456633261040381
82330 0.4105550172269493
39496 0.33280811450767495
...
```

```
R109 (Doc_ID BM25_Score) :
16953 1.131236307194757
26073 1.0479843111751264
61540 1.038283542199531
4933 0.9661551929278847
16575 0.904307070236808
64476 0.8730202244909762
```

```

67717 0.8497452282468656
78626 0.8418493584792328
34684 0.8112752050406455
23398 0.7921558872623953

```

...

```

R135 (Doc_ID BM25_Score) :
57876 0.6744874940228573
71894 0.5196381472811176
42848 0.5172924578571716
2738 0.40620267332577525
79698 0.030066821554749767
78998 0.030066821554749767
65311 0.03006577590333607
65501 0.030063613688166558
52721 0.030049043784249163
80891 0.030028313847379407

```

...

Task 4. Evaluate all three models using three standard effectiveness measures to provide a comprehensive comparison of ranking performance. Clearly report and analyse the results for each model across all datasets. In addition, describe the validation strategy used to tune the parameters of your custom model.

In this task, you are required to use the provided relevance judgments (Relevant_Judgements.zip) to evaluate the ranking outputs of three models located in the “RankingOutputs” folder. Specifically, you should apply the following three effectiveness measures to assess the document ranking performance based on your saved outputs.

- Average precision (and *MAP*),
- Precision at rank position 10 (precision@10) and their average, and
- Discounted cumulative gain at rank position 10 ($p = 10$), DCG_{10} (and *their average*)

$$DCG_p = rel_1 + \sum_{i=2}^p \frac{rel_i}{\log_2 i}$$

where $rel_i = 1$ if the document at position i is relevant; otherwise, it is zero.

- (a) Summarize your evaluation results in tabular form. Tables 1–3 provide example summary tables corresponding to the three effectiveness measures, respectively.

Table 1. The performance of 3 models on Average Precision

Topic	Baseline1	Baseline2	Model_C
R101	0.685	...	...
...		...	...
R109	0.635		
...			
R135	0.382		
...	...	...	...
R150	...	...	...
<i>MAP</i>	...	...	...

Table 2. The performance of 3 models on *precision@10*

Topic	Baseline1	Baseline2	Model_C
R101	0.5	...	...
...		...	...
R109	0.6		
...			
R135	0.2		
...	...	...	...
R150	...	...	...
<i>Average</i>	...	...	...

Table 3. The performance of 3 models on *DCG₁₀*

Topic	Baseline1	Baseline2	Model_C
R101	3.247	...	...
...		...	...
R109	3.707		
...			
R135	0.69		
...	...	...	...
R150	...	...	...
<i>Average</i>	...	...	...

- (b) Describe the parameters used in your custom model, clearly specifying their roles and their impact on ranking performance. Additionally, explain the validation process used to justify the selected parameter settings, including how empirical results support your choices.

Hint: *You may perform validation over the entire dataset collection and apply a parameter search method (e.g., grid search or systematic variation) to determine appropriate parameter values.*

Task 5. Conduct a statistical significance test to compare your custom model with two baseline models, and determine whether your model outperforms the baselines. You may use a t-test based on the evaluation results (e.g., those reported in Tables 1, 2, and 3). Specifically, perform pairwise comparisons between Baseline1 and Model_C, and between Baseline2 and Model_C.

Report the results of the t-test, including both the p-value and the t-statistic. The test can be conducted using a single effectiveness measure or multiple measures; using multiple measures generally provides stronger evidence against the null hypothesis. A common significance threshold is a p-value of 0.05.

Task 6. Application Scenario and Future Work

- (a) Describe a realistic application scenario in which your custom model could be deployed. Additionally, discuss relevant ethical considerations associated with the deployment. These may include, but are not limited to, data privacy, potential biases in training data, fairness across different user groups, or transparency & explainability of model decisions, etc.
- (b) Discuss potential directions for extending your custom model to further improve ranking performance and propose mitigation strategies for the ethical issues identified in (a).

Assignment 2 General Requirements

- The following are the frameworks/libraries that you could use for assignment 2:
 - (a) sklearn
 - (b) nltk
 - (c) pandas
 - (d) numPy
 - (e) Matplotlib
- If you want to use another package or library, you need to get your tutor's approval.
- You can re-use or update your assignment 1 code, the workshop solutions or review question solutions.
- Your programs should be well laid out, easy to read and well commented.
- All items submitted should be clearly labelled with your group number, name or student names.
- Marks will be awarded according to the marking guide (rubric). See the marking rubric for more details.
- You will lose marks for missing or inaccurate statements of completeness or user manual, and for missing sections, files, or items.
- We expect all team members to participate equally in this assessment project. If you have different contributions, you should provide the percentages and your signature on the cover page of the final report. If you have a major team conflict issue, your individual contributions will be assessed through peer review and tutor review.
- You can find “What you need to do”, “Final submission” and “Making Rubric” on Canvas.

END OF ASSIGNMENT 2